



DAFFODIL INTERNATIONAL UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
FYDP (PHASE-I) EVALUATION REPORT
REPORTING PERIOD- SUMMER 2026

Project Identification:

I. Project Title	An Explainable machine learning framework for university student dropout prediction and risk factor analysis	
II. Group Members	1. Name: Arnob Das Student ID: 0242310005101361 2. Name: Md Shazzadur Rahman Piash Student ID: 0242310005101583	
III. Supervisor	Name: Mr. Saiful Islam Designation: Assistant Professor	
IV. Co-Supervisor	Name: Md Tanvirul Islam Designation: Lecturer	
V. Submission Date:		
VI. Certificate:	“This is to certify that the final year design project work until Phase-I evaluation held on _____, titled as stated in <i>Sec. I</i>, executed by the students’ group mentioned in <i>Sec. II</i>, have been found satisfactory and every section of this report is reflecting the same.”	 (Signature of Supervisor & date)

Project Insights

Thematic Area(s): <i>[Just click the check box]</i>	Artificial Intelligence and Machine Learning	☒
	Deep Learning	☒
	Health Informatics	☐
	Cybersecurity	☐
	Software Engineering and Development	☐
	Blockchain Technology	☐
	Internet of Things (IoT)	☐
	Computer Networks	☐
	Computer Vision	☐
	Natural Language Processing (NLP)	☐
	Robotics	☐
	Game Development	☐
	Cloud Computing	☐
	Image Processing	☐
Others (please specify):		
Software packages, tools, and programming languages	[list down the software, programming languages, and packages you have used to do your project]	

CO Description for FYDP-Phase-I

CO	CO Descriptions	PO
CO4	Perform economic evaluation, cost estimation, and apply suitable project management procedures throughout the FYDP lifecycle in the context of developing the “An Explainable machine learning framework for university student dropout prediction and risk factor analysis ” Project.	PO11
CO6	Select and apply appropriate methodologies, resources, and contemporary engineering/IT tools for prediction, modeling, and solving complex engineering processes for the “An Explainable machine learning framework for university student dropout prediction and risk factor analysis” Project.	PO5
CO7	Assess societal, health, safety, legal, and cultural issues and responsibilities in professional engineering practice related to the FYDP problem.	PO6
CO10	Operate effectively as an individual and as a member/leader in multidisciplinary teams during FYDP.	PO9

1. Project Overview

1.1 Introduction

This research addresses a critical flaw in current dropout prediction systems: they predict but do not explain. Universities receive at-risk lists without understanding whether problems stem from attendance, academics, or finances. Without explanations, interventions become guesswork. This project combines LSTM for temporal modeling with SHAP for interpretability, creating a transparent system that tracks evolving risk while explaining each prediction. Pre-admission data enables earlier identification, and threshold analysis reveals when intervention becomes critical.

1.2 Background Study

Student dropout prediction has increasingly utilized statistical analysis, machine learning, and data analytics; however, most existing works struggle to balance temporal pattern recognition with model transparency. Early studies utilized regression models and ROC AUC analysis to forecast student dropout based on registration and demographic data [2]. Other statistical approaches have heavily emphasized sociodemographic and financial variables. For instance, event history analysis and multinomial logit models have been extensively applied to identify that minority status, lower-income, and being a first-generation student significantly increase dropout risks, while high GPA and strong motivation lead to timely graduation [14], [15], [16], [17]. While these statistical methods provide excellent population-level insights into financial and racial factors, they lack the individualized, dynamic predictive capability of modern machine learning.

Moving towards machine learning, previous research explored the early prediction of student dropout using standard algorithms like Random Forest and Decision Trees [8]. Large-scale predictive modeling has also been conducted using diverse global datasets. For example, traditional classifiers like Random Forest, Elastic Net, and CatBoost have been utilized across various cohorts—from 50,095 students in the US [11] to university students in Hungary [12] and South Korea [13]. These models successfully predicted dropouts achieving precisions up to 83% by highlighting static features such as high school GPA, behavioral metrics (e.g., campus event attendance), and tuition fees. However, a major limitation of these studies is their reliance on static, single-snapshot data, failing to capture the temporal changes in student behavior over successive semesters. Other researchers focused on specific platforms, predicting dropouts using Moodle activity trends [7] or regional Uwezo datasets in developing countries [4], which often lack a holistic view of a student's risk profile or the integration of sequential algorithms.

To address the need for sequential data processing, Deep Neural Networks (DNN) were applied on online learning platform data [9], analyzing video watch times to reach high accuracy. Further advancements proposed a Long Short-Term Memory (LSTM) network to forecast dropouts using multifaceted data [1], successfully identifying noticeable clustering patterns. The critical drawback of both of these deep learning frameworks is that they act as "black boxes" and completely lack transparency or explainability (XAI) in their decision-making processes.

To introduce interpretability into educational machine learning, Explainable AI (XAI) was applied to predict students' adaptability [6]. Similarly, XGBoost paired with SHAP was utilized to assess dropout risk using socioeconomic status and attendance [10]. While these achieved good interpretability, their major limitation lies in focusing on static, single-semester features rather than tracking dynamic, multi-semester temporal data. Furthermore, XAI-based personalized interventions were advocated [5], but predominantly relied on static interpretable models rather than deep sequential ones.

1.3 Gap Analysis

Reference Title	Authors	Publication Year	Dataset & Size	Methodology / Algorithms	Key Features / Inputs	Performance Metrics	Explainability (XAI)	Limitations / Future Work
[7] Long Short-Term Memory (LSTM) Based Model for Forecasting of Students' Dropout	Ibrahim Yunusa Umar, K.I. Musa, Muhammad Talla	2025	Student data (attendance, GPA, admission grade, GDP, unemployment)	LSTM Network	Sequence input layer, LSTM layer, fully connected layer	High accuracy, noticeable clustering patterns for target classes	Not specified / None	Did not explore explainability (XAI) for predictions
[2] Forecasting Student Dropout (Analysis of Regression and Demographics)	Johannes Benens, Karolin Schneider, Simon Gortz, S. Oetler, J. Burghoff	2019	Large dataset (graduates ~1000+, dropouts ~1000+ per semester)	Regression models, ROC AUC analysis	Semester-wise enrollment, performance over 4 semesters	Accuracy ranges from 83.47% (1st sem) to 93.76% (4th sem)	None	Static modeling; requires multi-semester tracking for high accuracy
[3] Big Data and Analytics / Prediction of student's performance	Zohar, L.M.A. et al. (Applied-11-00237-43)	2020	Small dataset size modeling / Survey based	Educational Data Mining, Prediction Models	Indirect assessments, major types	N/A	None	Focuses on small dataset size and general analytics
[4] Addressing student dropout in developing countries. Predicting Student Dropouts using High School and Entrance Exam Data	Nasima Mubina	2023	Uwezo (Tanzania, 61,340 samples) & India dataset (11,257 samples)	Machine Learning for highly imbalanced data	Pupil-Teacher Ratio, Income, Parent Interaction, Latrine ratio	Dealt with high imbalance (1.6% to 4.5% dropout rate)	None	Highly imbalanced data, needed heavy pre-processing and Imputation
[5] Predicting Student Dropouts using High School and Entrance Exam Data	Marcell Nagy, Richard McKinley	2023	Sensitive University Data	Machine Learning Models	Math scores, Chosen subject score, High school GPA, Language cert, Gender	Evaluated based on feature importance and average absolute impact	Feature importance tracking	Data is highly sensitive and restricted
[8] Prediction of Students' Adaptability Using Explainable AI in Educational Machine Learning Models	Leonard Chubawakaka Nnadi, Yusaku Watanabe, et al.	2023	Kaggle Dataset (Students' adaptability level in online education)	Educational Machine Learning Models + XAI	Demographics, Network type, Device, Financial condition	High accuracy in adaptability prediction	Yes (Explainable AI used)	Focused mainly on online education adaptability rather than absolute dropout
[7] Predicting student dropout using Moodle activity trends	M. Veerama and H. U	2024	University student records (Moodle logs, demographics)	Predictive Modeling with normalized variables	Moodle activity trend, Months absent, Accumulated credits, Failed courses	Calculated Spearman correlations	None	Dependent heavily on active Moodle usage logs
[6] Early Prediction of Student Dropout based on Machine Learning	Carlos Kemper et al.	2020	University student records (15k+ records)	Random Forest and Decision Trees	Academic performance, Demographics	Accuracy ~88%	None	Does not capture temporal changes in student behavior
[9] Student Dropout Prediction using Deep Learning Techniques	S. Kim et al.	2022	Online learning platform data	Deep Neural Network (DNN)	Video watch time, Quiz scores, Assignment completion	Accuracy ~91%	None	Lack of transparency in the model's decision making
[10] Interpretable Machine Learning for Student Dropout Risk Assessment	A. Sharma and P. Kumar	2023	Higher education dataset (5k+ records)	XGBoost with SHAP	Socioeconomic status, First-semester grades, Attendance	ROC-AUC 0.89	Used SHAP for feature contributions	Focused on static features rather than dynamic multi-semester data

The comparative analysis of the existing literature highlights a significant over-reliance on static data analysis for student dropout prediction. Most existing studies primarily depend on static machine learning models, such as Random Forest, XGBoost, and Logistic Regression, using cross-sectional data. While these conventional models provide reasonable predictive accuracy based on a single snapshot of information, they completely fail to track the dynamic, semester-by-semester academic progression of a student. Consequently, they are unable to analyze the data as a continuous sequence of academic events, missing the crucial temporal changes in student behavior over time.

To address the need for sequential data processing, some recent studies have successfully applied deep learning frameworks, specifically Long Short-Term Memory (LSTM) networks, to capture temporal dependencies. However, these advanced temporal models inherently function as "black boxes," failing to offer any interpretability or transparent explanation regarding why a particular student is classified as at-risk. On the other hand, while Explainable AI (XAI) tools like SHAP have been introduced to educational data mining, their application remains strictly restricted to static, non-sequential models. There is a distinct absence of research that effectively integrates robust interpretability with dynamic, multi-semester tracking.

Our proposed system explicitly addresses these identified research gaps by developing an integrated framework. By utilizing an LSTM network, the model effectively captures the long-term temporal dependencies of a student's academic journey across multiple semesters. Simultaneously, by integrating SHAP (SHapley Additive exPlanations), we crack open the "black box" of the LSTM architecture to provide transparent, personalized explanations for every single prediction. This unique combination of dynamic sequence tracking and robust individual interpretability provides a comprehensive and actionable solution that is largely missing in the current educational data mining literature.

2. Objectives

The objectives of Phase I are:

- I. To identify the key demographic, academic, and financial features available in the dataset for dropout prediction.
- II. To analyze the distribution, quality, and relationships of the dataset features to support effective model development.
- III. To preprocess the dataset by preparing the features for machine learning, including encoding and train-test splitting.
- IV. To develop and evaluate a baseline **Random Forest** classification model for predicting student dropout.

3. Methodology/ Requirement Specification:

3.1 Research Design/ Prototype Design

RESEARCH DESIGN / METHODOLOGY (PHASE I)

Student Dropout Prediction – Baseline Model Development

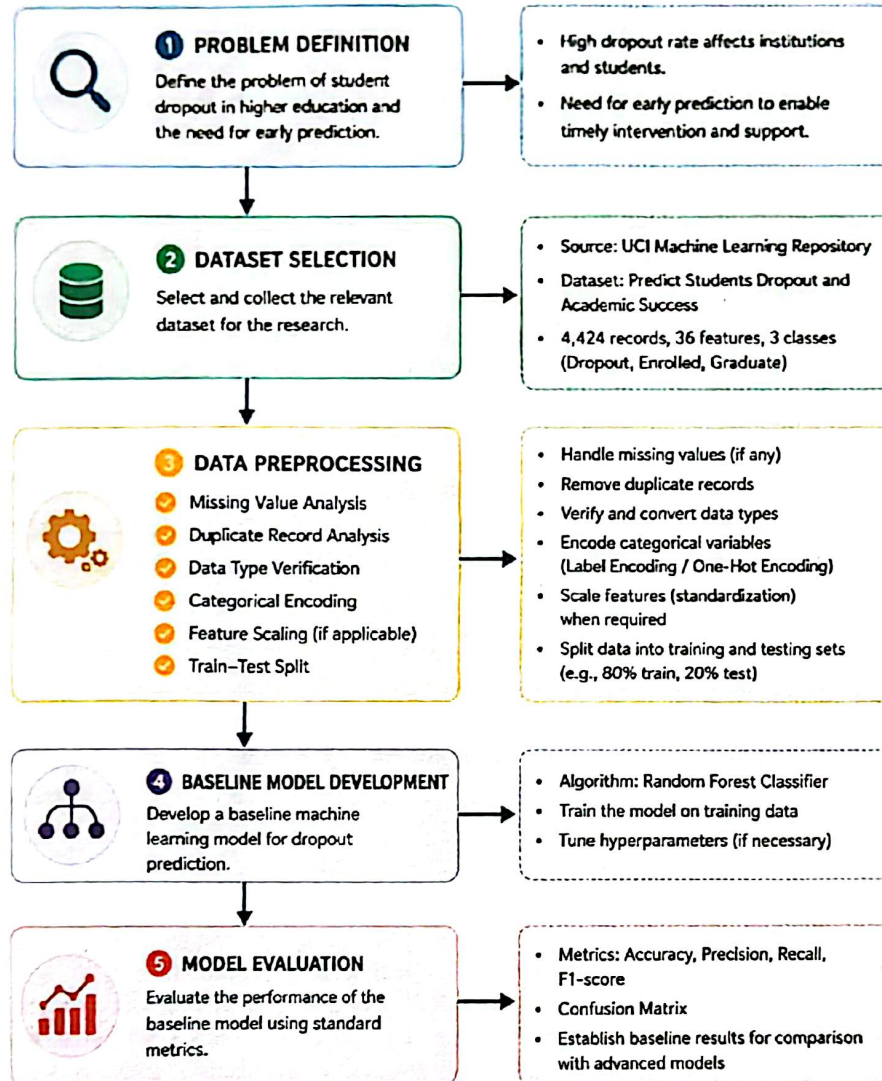


Figure 1: phase 1 methodology design

This research adopts a quantitative, data-driven methodology to develop a machine learning model for predicting university student dropout. The study utilizes the publicly available "Predict Students Dropout and Academic Success" dataset obtained from the UCI Machine Learning Repository.

The research process began with dataset selection and understanding, followed by Exploratory Data Analysis (EDA) to examine the dataset's characteristics, feature distributions, and class

balance. Based on the EDA findings, the dataset was preprocessed by encoding categorical variables and splitting the data into training and testing sets.

A Random Forest classifier was then developed as the baseline machine learning model. The model was trained using the processed dataset and evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and confusion matrix. The results obtained in this phase establish a baseline performance for comparison with more advanced models that will be implemented in subsequent phases of the research.

3.2 Data Collection/ Need Assessment

The dataset used in this research was obtained from the UCI Machine Learning Repository. It is a publicly available and licensed dataset titled "Predict Students Dropout and Academic Success", created from a higher education institution by integrating data from several institutional databases. The dataset contains 4,424 student records with 36 input features and one target variable representing three classes: Dropout, Enrolled, and Graduate.

The collected data includes students' demographic information, academic background, socioeconomic status, financial information, and academic performance during the first and second semesters. Since the dataset was already preprocessed by its creators, it contains no missing values or significant anomalies, making it suitable for machine learning analysis.

The dataset was selected because it provides comprehensive information relevant to student dropout prediction and includes semester-wise academic performance, which aligns with the objectives of this research. Python was used as the primary tool for loading, inspecting, and processing the dataset during the Exploratory Data Analysis (EDA) and baseline model development phases.

3.3 Analysis Techniques

The analysis was conducted using Python and its data science libraries, including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. Initially, Exploratory Data Analysis (EDA) was performed to understand the dataset structure, identify feature types, examine class distribution, detect duplicate records, and analyze relationships among variables using descriptive statistics and visualization techniques.

After EDA, the dataset was preprocessed by encoding categorical variables and splitting the data into training and testing sets. A Random Forest classifier was then developed as the baseline machine learning model to predict student dropout. The model's performance was evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and confusion matrix. The results obtained in this phase will serve as the baseline for comparison with more advanced models in the subsequent phases of the research.

4. Progress Achieved:

4.1 Completed Tasks

During this reporting period, the following tasks and milestones were successfully completed:

- I. **Literature Review:** Collected and reviewed relevant research papers on student dropout prediction, machine learning, and explainable artificial intelligence to understand existing approaches and identify research gaps.
- II. **Research Planning:** Finalized the research objectives, research questions, methodology, and selected the project title with supervisor approval.
- III. **Dataset Selection:** Selected the "Predict Students Dropout and Academic Success" dataset from the UCI Machine Learning Repository as the primary dataset for this research.
- IV. **Dataset Understanding:** Studied the dataset structure, feature descriptions, and target classes. The dataset features were categorized into demographic, academic, financial, and *socioeconomic factors*.
- V. **Exploratory Data Analysis (EDA):** Performed exploratory data analysis to examine the dataset characteristics, feature distributions, class distribution, descriptive statistics, duplicate records, and correlations among variables.
- VI. **Data Preprocessing:** Prepared the dataset for machine learning by encoding categorical variables and splitting the dataset into training and testing sets.
- VII. **Baseline Model Development:** Developed and evaluated a Random Forest classifier as the baseline prediction model. The evaluation results obtained in this phase will serve as a benchmark for comparison with advanced models in subsequent phases.

4.2 Results Obtained

The initial phase of the project has been successfully completed. The student dropout dataset was collected, cleaned, and analyzed through Exploratory Data Analysis (EDA). The dataset contains 4,424 student records with 36 input features and one target variable representing three student outcomes: Dropout, Enrolled, and Graduate. No missing values or duplicate records were found, indicating that the dataset is suitable for machine learning.

Several visualizations, including class distribution, correlation heatmap, histograms, and box plots, were generated to understand the characteristics of the data. The analysis showed that the dataset has a moderate class imbalance, with Graduate being the largest class and Enrolled the smallest. The visualizations also indicated that semester academic performance features exhibit noticeable differences across the target classes.

A baseline Random Forest classification model was trained using all available input features. The model achieved an accuracy of 76.72% on the test dataset. Feature importance analysis revealed that Curricular Units (2nd Semester Approved), Curricular Units (2nd Semester Grade), Curricular Units (1st Semester Approved), Curricular Units (1st Semester Grade), Admission

Grade, and Age at Enrollment were among the most influential features for predicting student outcomes.

These preliminary results demonstrate that the dataset is appropriate for the student dropout prediction task and provide a strong foundation for the next phase, which will involve feature selection, model comparison, and performance optimization.

5. Challenges Faced:

SI No.	Issues and Challenges	Strategies or Plans
1	Selecting an appropriate student dropout dataset with sufficient features for machine learning.	Reviewed multiple publicly available datasets and selected a well-structured dataset suitable for the research objectives.
2	Understanding the dataset attributes and relationships <i>between features</i> .	Conducted Exploratory Data Analysis (EDA) using statistical summaries and visualizations to <i>gain insights into the data</i> .
3	Setting up the Python environment and installing the required libraries for data analysis and model development.	Configured the development environment, installed the necessary packages, and verified the setup before proceeding.
4	Identifying the most relevant features for predicting student dropout.	Trained a baseline Random Forest model and used feature importance analysis to identify influential features. Further feature selection will be performed in the next phase.
5	Selecting an appropriate baseline machine learning model for initial evaluation.	Implemented a Random Forest classifier to establish baseline performance. Additional machine learning models will be compared in the next phase to identify the best-performing model.

6. Next Steps:

Outline the tasks and milestones planned for the next phase of the project.

SI No.	Next Task	Estimated Completion Time (MM-YY)
1	Perform feature selection and identify the most relevant features for student dropout prediction.	08-26
2	Train and compare multiple machine learning models (e.g., Logistic Regression, Decision Tree, SVM, Random Forest, XGBoost, The proposed LSTM).	09-26

3	Apply Explainable AI (XAI) techniques, such as SHAP (SHapley Additive exPlanations), to interpret model predictions and identify the most influential features and optimize the best-performing model using hyperparameter tuning and cross-validation.	10-26
4	Evaluate the final model using performance metrics such as Accuracy, Precision, Recall, F1-Score, ROC Curve, and Confusion Matrix.	10-26
5	Analyze the final results, prepare visualizations, complete documentation, and finalize the project report.	11-26

7. Updated Timeline:

Provide an updated timeline, highlighting progress made and indicating any adjustments.

Tasks	Weeks																
	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Dataset collection and selection + feature analysis																	
Background study + Gap analysis																	
EDA analysis																	
Baseline Model Training(Random Forest)																	

Estimated Work Period	
Actual Work Period	

8. Resources Utilized:

Following resources were utilized during the reporting period :

SI No.	Resource Type	Resources Utilized
1	Programming Language	Python 3.11

2	Development Environment	Visual Studio Code (VS Code) with Jupyter Notebook
3	Libraries	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
4	Dataset	Student Dropout and Academic Success Dataset
5	Version Control	Git and GitHub
6	Documentation	Research papers and related literature on student dropout prediction and Explainable AI (XAI)
7	Hardware	Personal laptop (Windows 11)

9. Project Management and Financial Analysis:

Project Management: The project is being managed using a phased development approach. Phase 1 focused on literature review, dataset collection, exploratory data analysis (EDA), and developing a baseline machine learning model. Future phases will include feature selection, model comparison, explainability (SHAP), evaluation, and documentation. Progress is monitored through regular meetings with the project supervisor.

Estimated Cost: The project requires minimal financial expenditure as it utilizes open-source software and publicly available datasets. The estimated cost is limited to electricity and internet usage during development. Existing personal computing resources are used, so no additional hardware or software purchases are required.

10. Future Considerations:

The following factors may impact the next phase of the project and will require careful consideration:

- I. **Model Selection:** Choosing the most suitable machine learning model to achieve high prediction accuracy.
- II. **Feature Selection:** Identifying the most relevant features while removing less significant ones without affecting model performance.
- III. **Model Optimization:** Applying hyperparameter tuning and cross-validation to improve the performance and generalization of the selected model.
- IV. **Explainable AI (SHAP):** Implementing SHAP to interpret the model's predictions and identify the most influential features affecting student dropout.

- V. **Model Evaluation:** Comparing multiple machine learning models using performance metrics such as Accuracy, Precision, Recall, and F1-Score to select the best-performing model.
- VI. **Final Documentation:** Completing the implementation, analyzing the results, and preparing the final project report and presentation.

11. Conclusion:

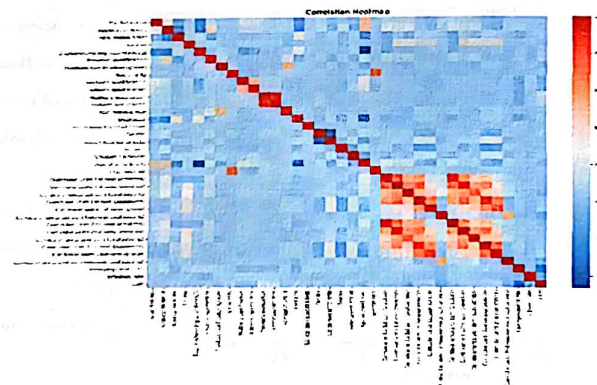
This progress report summarizes the work completed during the first phase of the Student Dropout Prediction project. The literature review, dataset collection, and Exploratory Data Analysis (EDA) were successfully completed, providing a clear understanding of the dataset and its characteristics. A baseline Random Forest model was developed, achieving an accuracy of **76.72%**, and feature importance analysis identified the most influential variables for prediction. The completed work provides a solid foundation for the next phase, which will focus on feature selection, model comparison, Explainable AI (SHAP), model optimization, and final evaluation to develop an accurate and interpretable student dropout prediction system.

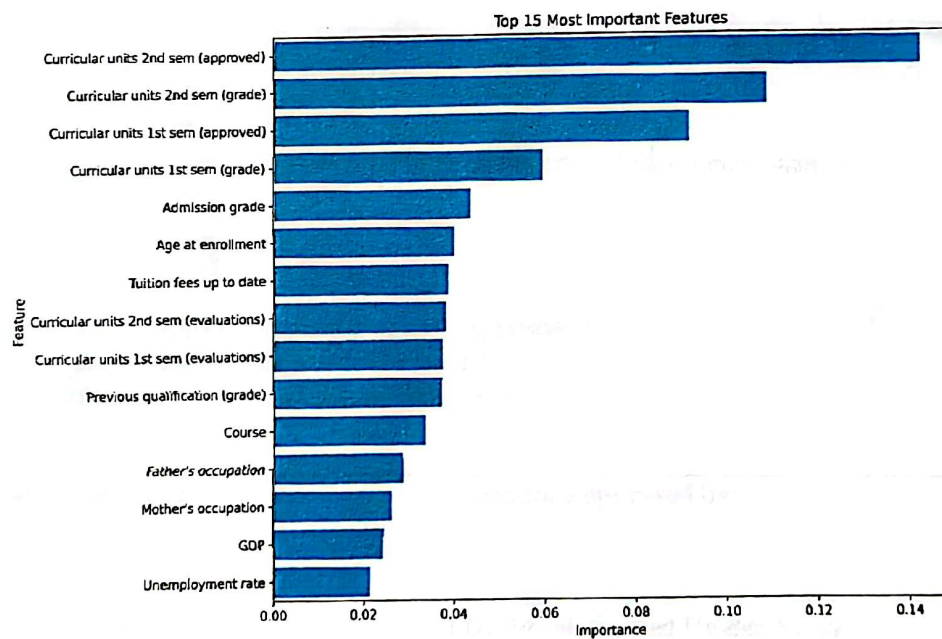
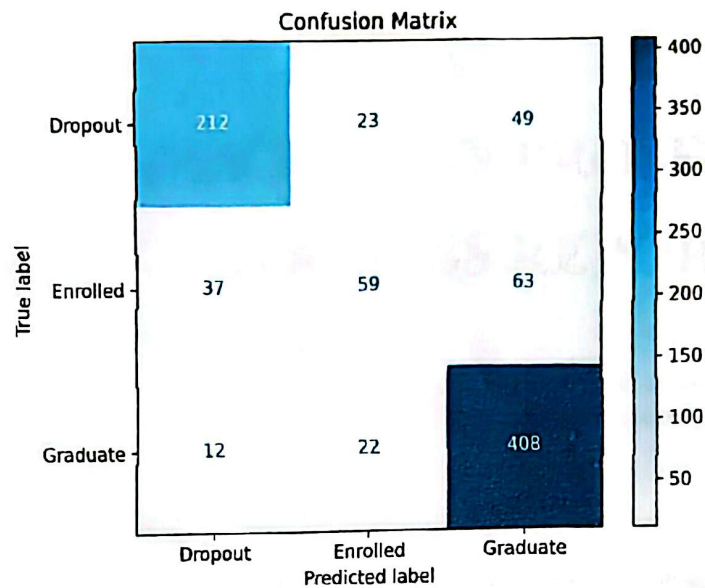
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Appendix:





GitHub Repository Link: <https://github.com/code-silence/student-dropout-prediction-research>

FINAL YEAR DESIGN PROJECT

PHASE-I PROGRESS REPORT

This report, in the form of a template, has been specifically designed for BSc. students working on their Final Year Design Project (FYDP) at Computer Science and Engineering Department, Daffodil International University (DIU).

Every group of students is required to do the following:

1. Complete all the sections of this template
2. Get it certified by the assigned supervisor before one week of Phase-I evaluation presentations
3. Submit 01 photocopy to each of the following, on or before the day of Phase-I presentations:
 - a. Supervisor
 - b. Internal Evaluator
4. Submit original copy to FYDP committee on the day of Phase-I presentations.

Note:

1. Use English
2. There should be NO grammatical or spelling mistakes
3. Submission after due date will not be accepted
4. For more information, contact your Supervisor

Template prepared by:	Template approved by:
FYDP Committee Dept. of CSE, DIU	Dr. Sheak Rashed Haider Noori Professor and Head, Dept. of CSE, DIU

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